

Substrate Pre-Screening, Independent Moderator Pathway, and Phantom Brand Persistence Phase B Extension

AIAS™ Presence Measurement Protocol, v1.6 — Three Methodology Increments under
Retrospective Test against the v0.16–v0.21 Corpus

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Abstract

The AIAS™ Presence Measurement Protocol v1.5 (SSRN 6810758) closed the within-cell adequacy and two-channel Recall layers of the multi-component construct. Two prospective phases under v1.5 — v0.20 (skin-care, SSRN 6811441) and v0.21 (cosmetics, SSRN 6815378) — surfaced three open items that the v1.5 routing matrix could not resolve at the methodology layer. First, v0.21 produced uniform-saturation C2 failure (distinct C_P count = 1 in all three cells; modal share 1.000 each); v0.20 produced differential C2 failure (distinct = 2 in Cell A, 3 in Cell B, 1 in Cell C). Under v1.5 both routed to FALSIFIED, collapsing two structurally distinct substrate states to one verdict. Second, the moderator hypothesis $H_{IdentityLoad_moderator}$ returned NARROWED in both phases despite increasingly strong substantive channel-asymmetry signals (v0.21 Cell B R_{cult} -lead of 2.83×, the program's strongest); the five-leg joint structure was pessimistic in the presence of Regime 4 falsification regardless of within-phase signal direction. Third, v0.21 surfaced strong Phantom Brand Persistence — Glossier (a v0.20 Cell B panel brand, not on the v0.21 panel) appeared in cultural-channel frames for all six panel LLMs in q6 — as a case-study observation but not a measured component of the construct.

Protocol v1.6 introduces three methodology increments addressing these items. Increment 1 is a substrate-level Recognition pre-screen: a substrate with distinct C_P count = 1 in all three cells is classified as uniform-saturation and routes to a fifth Regime 4 verdict state, REGIME-4-UNAVAILABLE-AT-RECOGNITION, distinct from FALSIFIED. Increment 2 introduces $H_{IdentityLoad_Direct}$, a parallel hypothesis evaluable independently of Regime 4's C2 conditions, scored per cell via $\delta = \bar{R}_{cult} - \bar{R}_{cat}$ with bootstrap percentile confidence intervals ($n = 10,000$; per-cell resampling with replacement; methodology-level seed locked for retrospective scoring at 1621). The verdict matrix is anchored to the IL-gradient prediction with a monotonic-gradient check on Cell C. Increment 3 lifts Phantom Brand Persistence from a descriptive observation to a measured component: $R_{phantom}$ counts off-panel reference-vocabulary mentions across the 36 Phase B responses against a substrate-specific vocabulary constructed as the union of the panel, a pre-registered top-50 market-share list, and prior-phase Phase B emergents cross-validated by the market-share list. Persistence threshold $K = 6$.

Retrospective scoring against the v0.16–v0.21 corpus is reported in §5. The headline finding is v0.21: Increment 1 classifies the substrate as uniform-saturation (the rule's first measured trigger); Increment 2 returns CONFIRMED for $H_{IdentityLoad_Direct}$ (Cell A $\delta = -3.13$, 95% CI excludes 0; Cell B $\delta = +7.12$, 95% CI excludes 0; Cell C $\delta = +1.00$ satisfies the monotonic-gradient check); Increment 3 returns CONFIRMED for $H_{PhantomBrandPersistence}$ with six off-panel brands clearing $K = 6$ (Urban Decay 13, Estée Lauder 13, Glossier 12, Too Faced 9, Make Up For Ever 7, Clinique 6). The validity anchor (Glossier $R_{phantom} \geq 6$) passes with margin. The off-panel phantom cohort exhibits channel signature tracking the brands' Iden-

tity Load: Estée Lauder and Clinique appear in canonical-channel frames only ($R_{cult} = 0$); Glossier appears in cultural-channel frames only ($R_{cat} = 0$). The IL-gradient mechanism that produces panel-level channel asymmetry also produces off-panel phantom channel asymmetry — a finding for the AIAS™ 1.0 synthesis.

For v0.20, Increment 2 returns PARTIAL: Cell B $\delta = +3.75$ with CI excluding 0 supports the IL prediction in the high-IL cell, but Cell A $\delta = +0.75$ (CI overlaps 0) is against prediction; the skincare substrate's signal lives in Cell B vs. Cell C rather than the conventional Cell A vs. Cell B contrast, a substantive substrate finding documented in §3. For v0.16–v0.19, Increment 1 narrative classification (all differential or mixed) is feasible from published Phase A distributions; Increment 2 retrospective bootstrap is not feasible because two-channel R_{cult} decomposition was a v1.5 prospective increment first acquired in v0.20. Increment 3 retrospective scoring is locked to v0.21 only; the construct's prospective scope begins at v0.22.

The pre-registration discipline (tag v1.6-prereg-r1, deposited at osf.io/ec6wh/methodology/v1_6/) locks the three increment specifications before retrospective scoring is run.

1. Introduction

1.1 What v1.5 closed and what it did not

The AIAS™ Presence Measurement Protocol formalizes a measurement procedure for AI Availability — the brand-level probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context (González Castro, 2026a). Protocol v1.5 (González Castro, 2026e) closed two methodology layers: the within-cell adequacy condition C2, refined into a multi-statistic rule requiring at least 3 distinct C_P values per cell with modal share ≤ 0.625 ; and the two-channel Recall decomposition, splitting category Recall into R_{cat} (anchored by best/recommended/quality frames in Phase B) and R_{cult} (anchored by popular/celebrity/viral/cult frames). Prospective tests under v1.5 ran on v0.20 (skincare, González Castro, 2026f) and v0.21 (cosmetics, González Castro, 2026g).

v1.5 §6 flagged three open items at the methodology lockpoint and noted that addressing them would require a methodology successor. The three items, with the empirical evidence each accumulated through v0.21:

Substrate-shape information collapses under FALSIFIED. v0.20 produced differential C2 outcomes — Cell A bimodal in C_P , Cell B passing both statistics, Cell C saturated — and routed to FALSIFIED. v0.21 produced uniform C_P saturation across all three cells (every panel brand scored $C_P = 6/6$; distinct = 1, modal share = 1.000 per cell) and also routed to FALSIFIED. Both verdicts are correct under the v1.5 routing matrix, but they carry different information about the substrate. A differential-failure substrate has Recognition discrimination available but fails the within-cell variance condition for at least one cell; a uniform-saturation

substrate has no Recognition discrimination available anywhere. Collapsing both to FALSIFIED loses the substrate-shape signal.

Five-leg joint structure is pessimistic in the presence of substrate-wide Regime 4 failure. H_IdentityLoad_moderator is a joint hypothesis evaluated across phases sharing the IL-gradient panel design. v1.5 specifies that the joint backs off (NARROWED) when constituent legs falsify. v0.21 produced its v1.5 leg with H_Regime4 FALSIFIED-on-uniform-saturation, propagating NARROWED to the joint — despite the within-phase IL-gradient channel-asymmetry signal being the program’s strongest to date (Cell B R_{cult} -lead of 2.83×; Cell A opposite-direction R_{cat} -lead). The joint structure suppressed within-phase signal that the substrate had clearly surfaced. The mechanism is structural rather than substantive: the joint reads Regime 4 falsification as evidence against the moderator regardless of whether the falsification arose from a within-phase signal contradicting the moderator (an informative falsification) or from a substrate-wide Recognition ceiling unrelated to the moderator construct (an uninformative falsification for moderator purposes).

Phantom Brand Persistence sits at the case-study layer. v0.21 §4.5 documented Glossier (a v0.20 Cell B panel brand, not on the v0.21 panel) appearing in cultural-channel frames for all six panel LLMs in q6 and three of six in q4 and q5, with Estée Lauder (also not on the v0.21 panel) appearing in the canonical channel for all six models ($R_{cat} = 13/18$). The pattern is empirically substantial — the strongest single demonstration of off-panel brand persistence in the program — but the construct is not a measured component of AIAS. The case-study layer is unstable: it surfaces persistent off-panel brands when the acquisition logs are read with the construct in mind, but does not produce a falsifiable verdict.

1.2 What v1.6 closes

Protocol v1.6 introduces three increments addressing the three items. Each increment is specified and locked in §2–§4, retrospectively applied where feasible in §5, and entered into the methodology log in §7 with explicit prospective vs. retrospective application boundaries.

Increment 1 (§2) inserts a substrate-level Recognition pre-screen before C2 routing. A substrate with distinct C_P count = 1 in all three cells is classified as uniform-saturation and bypasses C2; the four cell-level Regime 4 verdicts collapse to a single substrate-level verdict, REGIME-4-UNAVAILABLE-AT-RECOGNITION, distinct from FALSIFIED. Substrates that fail the pre-screen rule remain on the standard v1.5 sequence.

Increment 2 (§3) introduces H_IdentityLoad_Direct, a parallel hypothesis evaluable independently of Regime 4. The test statistic is per-cell channel asymmetry $\delta = \bar{R}_{cult} - \bar{R}_{cat}$ with bootstrap percentile confidence intervals; the verdict matrix is anchored to the IL-gradient prediction with an explicit monotonic-gradient check on Cell C. H_IdentityLoad_moderator (the v1.5 Regime-4-dependent joint hypothesis) is retained in the program; the two are presented as a methodological pair with distinct

applicability conditions, not as a replacement.

Increment 3 (§4) operationalizes Phantom Brand Persistence as a measured component. $R_{phantom}$ counts off-panel reference-vocabulary mentions across the 36 Phase B responses. The reference vocabulary is constructed as the union of the panel, a pre-registered top-50 substrate market-share list, and prior-phase Phase B emergents cross-validated by the market-share list — a construction that breaks the endogeneity hazard inherent in discovery-driven vocabulary expansion. The persistence threshold is $K = 6$ mentions across 36 responses, anchored to the v0.21 Glossier demonstration (6/6 panel LLMs in a single Phase B query).

1.3 Retrospective vs. prospective application boundary

The increments apply retrospectively where the data supports them and prospectively from v0.22. Increment 1 applies retrospectively across v0.16–v0.21 via classification of published Phase A distributions; only v0.20 and v0.21 have the modern CSV format that supports automated scoring, but the classification rule is computable from any phase’s published C_P distribution. Increment 2 applies retrospectively only to v0.20 and v0.21 — the two-channel R_{cult} decomposition is a v1.5 prospective increment, first acquired at v0.20; v0.16–v0.19 used single-channel mention scoring and no defensible retrospective channel re-mapping is available. Increment 3 retrospective scope is locked to v0.21 only — the substrate where the construct is empirically motivated and where the v0.20 Phase B logs supply the cross-validated-emergent input.

The pre-registration discipline locks all three specifications, the verdict matrices, the bootstrap parameters, the reference vocabulary construction rule, the persistence threshold, and the retrospective vs. prospective boundary at git commit, tag v1.6-prereg-r1. Retrospective scoring follows the lock; methodology amendment, if required by validity-check failure, follows the same discipline.

2. Increment 1 — Substrate-level Recognition pre-screen

2.1 Motivation

v0.21 produced the canonical uniform-saturation pattern: every panel brand in every cell scored $C_P = 6/6$, yielding distinct C_P count = 1 and modal share = 1.000 in each of Cell A (prestige), Cell B (celebrity-DTC / cult), and Cell C (drugstore / mass). v0.20 produced the canonical differential pattern: Cell A bimodal (distinct = 2, modal share = 0.750), Cell B passing the C2 multi-statistic (distinct = 3, modal share = 0.625), Cell C saturated (distinct = 1, modal share = 1.000). Under v1.5 both phases routed to FALSIFIED at C2. The verdicts are correct under the routing matrix and the substrate-shape information is not preserved.

The information asymmetry matters for the multi-component construct. A differential-failure substrate has

Recognition discrimination available in some cells; the failure is local, the within-cell variance condition is the binding constraint, and the substrate remains a candidate for further Recognition-anchored analysis (Phase A pivot validation, cell-level dissociation classification, IL-gradient Recall analysis with Recognition as a moderator). A uniform-saturation substrate has no Recognition discrimination anywhere; the Recognition channel is maximally coded; downstream analysis must either rely on Recall alone or be reframed for the saturated substrate condition.

2.2 Classification rule (locked)

A substrate is classified as **uniform-saturation** if and only if the distinct C_P count equals 1 in all three cells. Otherwise the substrate is classified as **differential**.

The single rule is the strictest falsifiable form for the 6-LLM panel: distinct = 1 in a cell forces modal share = 1.000 (the modal value carries all six panel responses), so the rule collects only those substrates where every panel LLM agrees on every brand's Recognition status in every cell. The rule does not require all brands to be recognized — only that within each cell, every brand's C_P value is identical. A substrate where all brands in Cell A score $C_P = 0/6$ (none recognized) and all brands in Cell B score $C_P = 6/6$ (all recognized) would trigger the rule, although that configuration has not been observed in the corpus.

The rule rejects v0.20 unambiguously (Cell A distinct = 2, Cell B distinct = 3 — neither cell singleton) and triggers on v0.21 exactly. Across v0.16–v0.19, narrative classification from published Phase A distributions returns differential or mixed for all four phases, with no phase exhibiting all-cells distinct = 1.

2.3 Routing change and the fifth verdict state

Substrates classified as uniform-saturation bypass C2 routing. The four cell-level Regime 4 verdicts that would otherwise be resolved (per the v1.5 routing matrix) collapse to a single substrate-level verdict: REGIME-4-UNAVAILABLE-AT-RECOGNITION. This is a new state in the Regime 4 verdict vocabulary, distinct from FALSIFIED. The semantic distinction is:

- **FALSIFIED** — the substrate fails the C2 within-cell variance condition in one or more cells; Recognition discrimination is available somewhere but the variance condition is not met. The substrate is closed at Regime 4 but the failure is a within-cell adequacy problem.
- **REGIME-4-UNAVAILABLE-AT-RECOGNITION** — the substrate exhibits substrate-wide Recognition saturation; Recognition discrimination is not available at all; C2 is ill-defined as an adequacy condition because the discrimination it tests does not exist in the substrate.

Both states close Regime 4 for the substrate but carry different downstream information for the AIAS™ 1.0 synthesis. A FALSIFIED substrate may be revisited with a different panel construction; a uniform-saturation

substrate is informative about the substrate itself rather than about the panel.

Under uniform-saturation routing, the moderator hypothesis is evaluated via H_IdentityLoad_Direct (Increment 2) independently of the Regime 4 outcome. Type 1 / Type 2 quadrant population and lwachu-pattern dissociation cases are reported as descriptive findings in the phase paper but are not pass/fail verdict components under this routing.

2.4 Retrospective classification across v0.16–v0.21

Table 1 reports the Increment 1 classification for the six-phase corpus.

Phase	Substrate	Distinct C_P per cell (A, B, C)	Modal share per cell (A, B, C)	Classification
v0.16	Kitchen knives	(narrative: variance present)	(narrative)	differential
v0.17	Premium kitchenware	(narrative: variance plus inadequacy)	(narrative)	differential
v0.18	Indie fragrance	(narrative: variance present)	(narrative; worst cell 0.69)	differential
v0.19	Audiophile headphones	(narrative: 2-cell design; mixed)	(narrative; Cell B 0.875)	mixed
v0.20	Skincare	(2, 3, 1)	(0.750, 0.625, 1.000)	differential
v0.21	Cosmetics	(1, 1, 1)	(1.000, 1.000, 1.000)	uniform- saturation

v0.21 is the first measured trigger of the uniform-saturation rule. The classification entered the Regime 4 verdict under v1.6 as REGIME-4-UNAVAILABLE-AT-RECOGNITION; the v1.5 verdict under the same data was FALSIFIED.

3. Increment 2 — Independent moderator pathway

3.1 Motivation

H_IdentityLoad_moderator under v1.5 is a five-leg joint hypothesis: a CONFIRMED leg per phase requires the phase's H_Regime4 verdict to be CONFIRMED or PARTIAL and the within-phase IL signature to be mono-

tonic across cells. The joint backs off (NARROWED) when any leg falsifies. The structure correctly aggregates evidence across phases when each leg's H_Regime4 verdict is informative for the moderator question, but treats H_Regime4 falsification as evidence against the moderator regardless of the falsification's substantive content.

v0.21 surfaces this issue cleanly. The v0.21 leg has H_Regime4 FALSIFIED-on-uniform-saturation, propagating NARROWED to the joint. The within-phase IL signature, however, is the strongest in the program: mean R_{cat} values across cells are A = 5.75, B = 3.88, C = 2.00; mean R_{cult} values are A = 2.62, B = 11.00, C = 3.00. The channel-asymmetry estimate $\delta = \bar{R}_{cult} - \bar{R}_{cat}$ is -3.13 in Cell A, $+7.12$ in Cell B (a $2.83 \times R_{cult}$ lead — the program's strongest), $+1.00$ in Cell C — monotonic in the IL-predicted direction across all three cells. The joint structure suppressed this signal because the Recognition channel was saturated, not because the moderator construct was unsupported.

The fix is a parallel hypothesis evaluable on the channel-asymmetry signal directly, with confidence intervals that allow falsification at the within-phase layer rather than through joint-leg propagation.

3.2 Test specification (locked)

H_IdentityLoad_Direct is specified as follows. For each cell, compute $\delta = \bar{R}_{cult} - \bar{R}_{cat}$ over the n brands in the cell. Bootstrap a percentile confidence interval on δ with $n_{iter} = 10,000$, per-cell resampling with replacement, and confidence level 95%. The CI is reported as $[\delta_{lo}, \delta_{hi}]$ at the 2.5th and 97.5th percentiles of the bootstrap distribution of $\bar{R}_{cult} - \bar{R}_{cat}$ over the resampled brand-level draws.

Bootstrap reproducibility is locked at the methodology layer for retrospective scoring and at the phase layer for prospective scoring. For retrospective scoring against v0.16–v0.21, the methodology-level seed is `SEED_V16_RETRO = 1621`; per-cell sampling uses a deterministic offset (`SEED + ord(cell_id)`) so that all retrospective runs against the same data produce identical CIs. For prospective phases v0.22 and later, each phase pre-registration locks its own phase-level seed.

The CI method is percentile, not bias-corrected accelerated (BCa). Percentile CIs are locked because they are simpler to specify and produce stable behavior under the small n typical of cell-level resampling ($n = 8$ per cell pre-floor, $n \geq 5$ post-attrition under the program's adequacy rules). BCa is reserved for cases where the methodology paper documents specific need; the present specification does not require it.

3.3 Verdict matrix (locked)

The IL-gradient prediction is: Cell B (high-IL by panel design) shows $\delta > 0$; Cell A (medium-IL) shows $\delta < 0$; Cell C (low-IL) shows δ value bounded by the monotonic-gradient check. The monotonic-gradient check requires Cell C δ to lie between Cell A δ and Cell B δ , i.e., $\delta_A \leq \delta_C \leq \delta_B$ under the IL-predicted

ordering.

The verdict matrix is:

- **CONFIRMED** — Cell B $\delta > 0$ with CI excluding zero; Cell A $\delta < 0$ with CI excluding zero; Cell C monotonic-gradient check passes.
- **PARTIAL** — focal-cell directions match the IL prediction (Cell A $\delta < 0$ and Cell B $\delta > 0$) but at least one focal CI overlaps zero, OR all focal directions and CIs are intact but the Cell C monotonic-gradient check fails.
- **FALSIFIED** — Cell B $\delta \leq 0$ with CI excluding zero (opposite direction to IL prediction in the high-IL cell), OR Cell A $\delta \geq 0$ with CI excluding zero, OR both focal CIs exclude zero in the IL-predicted directions but Cell C breaks the monotonic-gradient check (a configuration that cannot be reconciled with the IL-gradient model).
- **INDETERMINATE** — brand-level post-attribution $n < 5$ in any focal cell (A or B). Cell C post-attribution shortfall does not trigger INDETERMINATE on its own; the gradient check defaults to “not evaluable” and the verdict resolves on focal cells only.

The verdict matrix preserves the v1.5 falsifiability discipline (Cell B in the wrong direction is the diagnostic falsification path) while opening a CONFIRMED path that v1.5’s joint structure did not permit when Regime 4 falsified on substrate grounds.

3.4 Retrospective application — v0.20 and v0.21

Table 2 reports the H_IdentityLoad_Direct verdict for v0.20 and v0.21 under the locked specification.

Phase	Cell A δ , CI	Cell B δ , CI	Cell C δ , CI	Gradient	Verdict
v0.20	+0.75, [−0.25, +2.12] (incl 0)	+3.75, [+2.25, +5.25] (excl 0)	−2.50, [−5.25, +0.38] (incl 0)	Cell A direction against IL prediction	PARTIAL
v0.21	−3.13, [−6.00, −0.25] (excl 0)	+7.12, [+4.75, +10.25] (excl 0)	+1.00, [−0.50, +3.25] (incl 0)	−3.13 ≤ +1.00 ≤ +7.12 passes	CONFIRMED

The v0.21 verdict is the program’s first CONFIRMED moderator outcome at any layer. The substrate signal is unambiguous: Cell B’s $\delta = +7.12$ with CI excluding zero by a wide margin (lower bound +4.75) directly evidences the IL-gradient prediction that brands with high consumer-discovery-channel cultural footprint (Rare Beauty, Fenty Beauty, Haus Labs, Pat McGrath Labs, Charlotte Tilbury, Huda Beauty, Kylie

Cosmetics, Anastasia Beverly Hills) are over-represented relative to canonical-channel frames in the cell's cultural-channel Phase B retrieval. Cell A's $\delta = -3.13$ with CI excluding zero directly evidences the opposite-direction prediction for medium-IL prestige brands (MAC Cosmetics, NARS, Bobbi Brown, Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille, Laura Mercier). Cell C's $\delta = +1.00$ satisfies the monotonic-gradient check.

The v0.20 verdict is PARTIAL with substrate-specific structure. Cell B's $\delta = +3.75$ with CI excluding zero supports the IL-gradient prediction in the high-IL cell; the direction and magnitude track the cosmetics result at smaller scale. Cell A's $\delta = +0.75$, however, is against the IL-prediction direction — heritage/clinical skincare brands (CeraVe, La Roche-Posay, Eucerin and the cell's other panel members) do not concentrate in canonical-channel frames the way cosmetics' Cell A prestige brands do. The CI overlaps zero ($[-0.25, +2.12]$), so the verdict path is PARTIAL rather than FALSIFIED. Cell C's $\delta = -2.50$ — substantial negative — carries the substrate's R_{cat} dominance: mass/drugstore skincare names (Cetaphil, Neutrogena, and the cell's other panel members) are canonical-channel-dominant. The skincare IL-gradient signal lives in Cell B vs. Cell C, not Cell A vs. Cell B as the standard prediction frames it. This is a substantive finding about the skincare substrate's category structure and is documented for the AIAS™ 1.0 synthesis: the IL-gradient model holds in cosmetics (the cultural-footprint cell is the high-IL cell as anticipated) but the canonical-channel pole shifts from prestige to mass under skincare's category structure.

3.5 Relationship to H_IdentityLoad_moderator

H_IdentityLoad_Direct does not replace H_IdentityLoad_moderator. The two hypotheses test different scopes: the Direct hypothesis tests within-phase channel asymmetry against the IL-gradient prediction; the moderator hypothesis tests the cross-phase consistency of the IL-gradient signal as a moderator of Regime 4 routing. Both stay in the program. Future phase pre-registrations may register either or both — the choice is locked at phase pre-reg per the program's standard discipline. Where both are registered, the joint moderator verdict can be evaluated under either H_IdentityLoad_Direct (the Increment 2 path) or H_IdentityLoad_moderator (the v1.5 Regime-4-dependent path); the v1.6 §5 retrospective scoring reports both for v0.20 and v0.21.

4. Increment 3 — Phantom Brand Persistence Phase B extension

4.1 Motivation

v0.21 §4.5 documented the strongest single Phantom Brand Persistence demonstration in the program. Glossier — a v0.20 Cell B panel brand, not on the v0.21 cosmetics panel — appeared in cultural-channel Phase B frames for all six panel LLMs in q6 (a 6/6 saturation event in a single query) and in three of six in

both q4 and q5. Estée Lauder — also not on the v0.21 panel — appeared in canonical-channel frames for all six models across all three q1–q3 frames, accumulating R_{cat} -equivalent = 13/18 mentions. Neither brand contributes to the panel-internal scoring, which is correct under the v1.5 methodology. But the persistent off-panel surface they produce is empirically substantial and was not captured by the protocol.

The construct is closely related to the v1.4 multi-component framing but operates one layer outward: panel-internal scoring measures whether named brands appear in the panel's retrieval; phantom scoring measures whether off-panel brands appear in the panel's retrieval with persistence. Within an LLM intermediary's category retrieval, persistent off-panel brand mention is itself a presence signal — the intermediary is treating an off-panel brand as a category member at the discovery layer. v1.6 lifts this from descriptive case-study reporting to a measured component.

4.2 Reference vocabulary construction (locked)

The reference vocabulary per substrate is the **union** of three exogenous sources:

1. **Panel.** The 24 panel brands for the substrate (3 cells $\times$ 8 brands).
2. **Top-50 market-share list.** A substrate-specific top-50 brands by market share, source pre-registered per substrate (e.g., for the cosmetics substrate, the source is a composite of publicly available cosmetics brand rankings — WWD Top 100 Beauty 2024, Statista Cosmetics Brands by Revenue Worldwide 2024, Allure Best of Beauty 2024 expanded brand index — with the list snapshot frozen at v1.6-prereg-r1 lock date). The citation is locked at phase pre-registration; the snapshot is frozen at acquisition.
3. **Cross-validated emergents.** Brands that appeared in prior-phase Phase B response logs for the substrate AND also appear in the top-50 market-share list. Brands found only in Phase B logs (without market-share corroboration) are excluded from the vocabulary.

The construction is exogenous in a specific sense: no brand enters the vocabulary purely through the surface being measured. The Phase B logs that motivated the construct (the v0.20 cosmetics-adjacent Phase B responses that produced Glossier as a candidate phantom) can contribute brands to the vocabulary only if those brands are also corroborated by an independent market-share list. The construction breaks the endogeneity hazard inherent in discovery-driven vocabulary expansion: a brand persistently surfacing in Phase B logs is not, by itself, evidence that the brand should be in the vocabulary against which Phase B mentions are subsequently scored. Glossier — the v0.21 anchor case — appears on every reasonable cosmetics market-share list independently of its v0.20 appearance, so the rule retains it; brands that surface only in Phase B logs without market-share corroboration are excluded.

Vocabulary size is variable per substrate (panel 24 plus market-share intersection plus cross-validated emergents typically yielding 50–80 brands). The vocabulary is held constant across the three cells within a sub-

strate; phantom scoring is substrate-level, not cell-level.

4.3 Measurement and verdict matrix

For each Phase B response, the full reference vocabulary is scanned for brand mentions using the v1.4 canonical mention detection rules (case-insensitive, accent-stripped, word-boundary anchored). Per off-panel brand, $R_{phantom}$ is the count of mentions across all 36 Phase B responses (max 36), decomposed as $R_{cat_phantom}$ (max 18, q1–q3 responses) and $R_{cult_phantom}$ (max 18, q4–q6 responses).

The persistence threshold is $K = 6$ mentions across the 36 Phase B responses. The threshold is anchored to the v0.21 Glossier demonstration: Glossier appeared in 6/6 panel LLMs in a single Phase B query (q6), producing a single-query 6-mention floor that constitutes the empirical anchor for the construct. Off-panel brands with $R_{phantom} \geq K$ are reported as phantom cases for the substrate.

The verdict matrix for **H_PhantomBrandPersistence** is:

- **CONFIRMED** — at least one off-panel reference-vocabulary brand has $R_{phantom} \geq K$.
- **ABSENT** — zero off-panel reference-vocabulary brands have $R_{phantom} \geq K$.

The construct’s calibration validity is anchored to the v0.21 retrospective scoring: Glossier $R_{phantom} \geq 6$ is required under v1.6 measurement. Failure of the validity anchor triggers a pre-registration amendment to revisit K, not silent re-tuning post-hoc. The retrospective scoring (§4.4) confirms the validity anchor with margin.

4.4 Retrospective application — v0.21

Table 3 reports v0.21 phantom scoring. Six off-panel reference-vocabulary brands cleared $K = 6$:

Brand	$R_{cat_phantom}$ (max 18)	$R_{cult_phantom}$ (max 18)	$R_{phantom}$ (max 36)	Channel signature
Urban Decay	8	5	13	mixed, slight canonical lean
Estée Lauder	13	0	13	pure canonical
Glossier	0	12	12	pure cultural (validity anchor)
Too Faced	3	6	9	cultural-leaning
Make Up For Ever	6	1	7	canonical-leaning
Clinique	6	0	6	pure canonical

The validity anchor passes: Glossier $R_{phantom} = 12$, required ≥ 6 . The verdict is **CONFIRMED** with substantial margin.

The off-panel phantom cohort exhibits a striking channel signature. Three of the six phantoms are channel-pure: Estée Lauder and Clinique appear in canonical-channel frames only ($R_{cult} = 0$ for both); Glossier appears in cultural-channel frames only ($R_{cat} = 0$). The channel signatures track the brands' Identity Load: Estée Lauder and Clinique are prestige/heritage in cosmetics terms (low-IL canonical brands by the program's IL-gradient framing); Glossier is celebrity/DTC/cult (high-IL cultural brand). The IL-gradient mechanism that produces panel-level channel asymmetry (formally tested as $H_{IdentityLoad_Direct}$ in §3) also produces off-panel phantom channel asymmetry. The Phantom Brand Persistence construct is not a separate phenomenon from IL-gradient channel asymmetry; it is the same mechanism extending past the panel boundary.

This finding has implications for the AIAS™ 1.0 synthesis (§6). The multi-component construct is currently specified panel-internally: C_P , R_{cat} , R_{cult} are all computed against the panel. The off-panel phantom layer reveals that the IL-gradient mechanism is substrate-level, not panel-level, and is recoverable through a vocabulary specification that breaks the endogeneity hazard. AIAS™ 1.0's foundational construct can incorporate the phantom layer as a measured component alongside the panel-internal layers without redefining the panel-anchored measurement procedure.

4.5 Retrospective application scope

The Phantom increment ships as **prospective-only from v0.22+**, with retrospective scoring on v0.21 only. The retrospective construction of reference vocabularies for v0.16–v0.20 is deferred — for v0.16–v0.18 the data formats predate the multi-channel Phase B decomposition and would not produce comparable $R_{cat_phantom} / R_{cult_phantom}$ values; for v0.19 the substrate (audiophile headphones) is empirically unmotivated for phantom analysis (the v1.6 motivation comes from cosmetics and skincare, not headphones); for v0.20 the data format is compatible but the cross-validated-emergents source is the substrate itself (a circularity that the locked construction rule was designed to avoid). The v0.22 prospective application is the natural next test, with the v0.21 vocabulary and threshold acting as calibration.

5. Joint retrospective scoring of v0.16–v0.21 under v1.6

Table 4 reports the joint retrospective scoring across the six-phase corpus.

Phase	Substrate	Routing (v1.5)	Routing (v1.6)	Moderator (v1.5)	Moderator (v1.6 — H_IL_Direct)	
					H_IL_Direct)	Phantom
v0.16	Kitchen knives	Standard (v1.4)	differential (narrative)	PARTIAL leg	N/A — channel data not collected	N/A — out of scope
v0.17	Premium kitchenware	Standard (v1.4)	differential (narrative)	FALSIFIED- on-panel- inadequacy	N/A — channel data not collected	N/A
v0.18	Indie fragrance	Standard (v1.4)	differential (narrative)	PARTIAL joint	N/A — channel data not collected	N/A
v0.19	Audiophile headphones	v1.4.x (2-cell design); C2 FAILED → UNDETER- MINED	mixed (narrative)	Excluded (uniform-IL design)	N/A — single- channel q1-q6 design	N/A
v0.20	Skincare	v1.5 full sequence; C2 FALSIFIED	differential	NARROWED (5-leg joint)	PARTIAL	N/A
v0.21	Cosmetics	v1.5 full sequence; C2 FALSIFIED	uniform- saturation → REGIME-4-UNAVAILABLE-AT-RECOGNITION	NARROWED (5-leg joint)	CONFIRMED	CONFIRMED

The before-and-after deltas:

v0.21. Regime 4 verdict moves from FALSIFIED to REGIME-4-UNAVAILABLE-AT-RECOGNITION. Moderator verdict moves from NARROWED to CONFIRMED. Phantom verdict moves from descriptive case study (v0.21 §4.5) to CONFIRMED with six off-panel brands clearing $K = 6$. The phase that produced the program's strongest within-phase IL-gradient signal now records that signal as a confirmed methodological finding rather than absorbing it into a backed-off joint.

v0.20. Moderator verdict moves from NARROWED to PARTIAL. The uplift is modest and accompanied by

a substantive finding: the skincare substrate’s IL-gradient signal lives in the Cell B vs. Cell C contrast rather than Cell A vs. Cell B. The Cell A canonical-channel pole expected for prestige/heritage substrates does not hold for skincare’s clinical/dermatologist Cell A. The PARTIAL verdict at the methodology layer is consistent with the substrate’s actual signal architecture.

v0.16–v0.19. Increment 1 narrative classification across these four phases returns differential or mixed for all four; no phase exhibits all-cells distinct $C_P = 1$. The narrative classification is consistent with the prospective application boundary documented in §7. Increment 2 retrospective scoring is not feasible for these phases — two-channel R_{cult} decomposition is a v1.5 prospective increment first acquired in v0.20 — and the §5 table records this as “channel data not collected” rather than as an INDETERMINATE verdict. Increment 3 is out of retrospective scope per §4.5.

The retrospective scoring is now the empirical anchor for the three increment specifications and for the prospective application boundary that begins at v0.22.

6. Implications

6.1 For v0.22+ prospective phases

The three increments enter the standard phase pre-registration discipline at v0.22 and after. Phase pre-registration locks: the substrate market-share source and snapshot date for reference vocabulary construction (Increment 3); the phase-level bootstrap seed for `H_IdentityLoad_Direct` (Increment 2); and the substrate-classification result, which can be computed deterministically from Phase A acquisition before C2 routing is invoked (Increment 1). The three increments do not change the panel construction discipline, the Phase A probe template, the Phase B six-frame battery, or the v1.5 multi-statistic C2 rule for substrates classified as differential. Phases under v1.6 inherit the v1.5 sequence unchanged for differential substrates and add the new routing for uniform-saturation substrates.

The verdict matrices add two new verdict states to the program’s vocabulary: `REGIME-4-UNAVAILABLE-AT-RECOGNITION` (Increment 1) and `H_PhantomBrandPersistence: CONFIRMED / ABSENT` (Increment 3). `H_IdentityLoad_Direct` adds verdict states `CONFIRMED / PARTIAL / FALSIFIED / INDETERMINATE` evaluable parallel to `H_IdentityLoad_moderator` (Increment 2). The verdict-state expansion is non-breaking — existing v1.5 verdicts retain their semantics — and is documented in §7.

6.2 For AIAS™ 1.0 synthesis

AIAS™ 1.0 is the program’s foundational construct synthesis. The five-substrate-family anchor base (kitchenware, fragrance, audio, skincare, cosmetics) is complete with v0.21. v1.6 tightens three methodology layers

before the synthesis:

The substrate-shape verdict distinction (Increment 1) lets the synthesis discuss uniform-saturation substrates as a separate category alongside differential-success and differential-failure substrates. Cosmetics is the first uniform-saturation case observed; the synthesis can treat it as informative about the cosmetics category itself (maximally Recognition-coded at the panel layer) rather than as an instance of methodology failure.

The independent moderator pathway (Increment 2) provides a within-phase falsifiable test of the IL-gradient prediction that does not require Regime 4 to clear. The synthesis can report $H_IdentityLoad_Direct$ verdicts per phase as the within-phase signal, distinct from the cross-phase moderator joint. The v0.21 CONFIRMED verdict becomes the strongest single piece of evidence for the IL-gradient construct in the program.

The phantom layer (Increment 3) extends the multi-component construct past the panel boundary while preserving the panel-anchored measurement discipline. The synthesis can incorporate phantom verdicts as a measured component alongside the panel-internal Recognition $\times$ Recall layers without redefining the construct. The channel-signature finding from v0.21 (§4.4) is a precursor for the synthesis: the IL-gradient mechanism operates substrate-wide, not panel-internally.

6.3 For the multi-component construct

v1.4 introduced the multi-component construct as Recognition $\times$ Recall with subsequent two-channel Recall decomposition in v1.5. v1.6 leaves the panel-internal multi-component specification unchanged but adds a substrate-level phantom layer that is conceptually distinct: panel-internal scoring measures the panel's behavior at Recognition and Recall layers; phantom scoring measures the panel's spontaneous import of off-panel brands into category retrieval. The two are complementary. Phases under v1.6 report both as standard outputs.

The construct's measured surface area expands from $C_P \times R_{cat} \times R_{cult}$ (panel-internal) to $\{C_P, R_{cat}, R_{cult}, R_{phantom}\}$ (panel-internal plus phantom-substrate-level). The channel decomposition extends to the phantom layer ($R_{cat_phantom}, R_{cult_phantom}$), preserving the v1.5 two-channel framing across the new component.

7. Methodology log

7.1 v1.5 $\rightarrow$ v1.6 increment trail

The methodology lineage from v1.2 to v1.6 is summarized below.

- **v1.2** (González Castro, 2026b) — construct validity notes; four-regime taxonomy.
- **v1.3** (González Castro, 2026c) — Phase A pivot-validation specification.
- **v1.4** (González Castro, 2026d) — Recognition $\times$ Recall decomposition; multi-component AI Availability.
- **v1.5** (González Castro, 2026e) — multi-statistic C2; two-channel Recall (R_{cat} , R_{cult}).
- **v1.6** (present paper) — substrate Recognition pre-screen; H_IdentityLoad_Direct; Phantom Brand Persistence Phase B extension.

Each increment paper has been a methodology successor rather than a revision: v1.5 did not modify v1.4's panel construction or probe template; v1.6 does not modify v1.5's C2 rule (for differential substrates) or the two-channel Recall decomposition. The program's measurement discipline is additive at the methodology layer; pre-registration locks at successive tags preserve falsifiability across the methodology trajectory.

7.2 Retrospective vs. prospective application boundary (locked)

Table 5 documents the application boundary for each v1.6 increment across phases.

Phase	Inc 1 (substrate pre-screen)	Inc 2 (H_IdentityLoad_Direct)	Inc 3 (Phantom)
v0.16	applied (retrospective, narrative)	not applied (R_{cult} unavailable)	not applied
v0.17	applied (retrospective, narrative)	not applied (R_{cult} unavailable)	not applied
v0.18	applied (retrospective, narrative)	not applied (R_{cult} unavailable)	not applied
v0.19	applied (retrospective, narrative)	not applied (R_{cult} unavailable; single-channel design)	not applied
v0.20	applied (retrospective, scored)	applied (retrospective, scored)	not applied (cross-validated-emergent source is the substrate itself)
v0.21	applied (retrospective, scored — first uniform-saturation trigger)	applied (retrospective, scored — first CONFIRMED)	applied (retrospective, scored — calibration anchor)
v0.22+	prospective	prospective	prospective

	Inc 1 (substrate pre-screen)	Inc 2 (H_IdentityLoad_Direct)	Inc 3 (Phantom)
Phase			

The “not applied” entries are documented as data-availability constraints, not as INDETERMINATE verdicts. The retrospective vs. prospective boundary is locked at v1.6-prereg-r1 and is not subject to post-hoc revision; future methodology successors may extend retrospective coverage but must do so under their own pre-registration discipline.

7.3 Pre-registration trajectory for v0.22+

v0.22 will be the first phase pre-registered under v1.6. The phase pre-registration will lock, at minimum: substrate; cell structure with IL-gradient stratification (or alternative design with explicit motivation); panel construction; Phase A probe template; Phase B six-frame battery; substrate market-share source citation for Increment 3 reference vocabulary; phase-level bootstrap seed for Increment 2; substrate classification expected outcome (computable ex-ante from Phase A acquisition); and phase hypotheses with verdict matrices. The standard pre-registration discipline (git commit, tag v0.NN-prereg-rN, OSF deposit) is unchanged.

The v0.22 substrate is not specified in the present paper and is reserved for the v0.22 pre-registration round.

Declarations

Funder. Self-funded.

Ethics. Not applicable; no human subjects.

Conflict of interest. The author is employed by Samsung Electronics America (Director, Corporate Brand Creative & Governance). The pre-registration discipline for the AIAS™ measurement program includes a per-phase ex-ante COI screen confirming no brand in the panel is Samsung-affiliated (documented as DEVIATIONS Entry 0 in each phase pre-registration). The v1.6 methodology paper introduces no new panel; the COI screen carries forward from the empirical anchor phases. The employer has no role in the research program’s methodology, pre-registration, scoring, or publication decisions.

Open data and pre-registration. All pre-registrations, registries, scoring scripts, retrospective scoring outputs, and reference vocabularies are deposited at osf.io/ec6wh/methodology/v1_6/. The v1.6 pre-registration is locked at git commit, tag v1.6-prereg-r1. Reference vocabulary for the v0.21 retrospective Phantom scoring is at methodology/v1_6/reference_vocabs/cosmetics_v021.json.

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